

Senior AI Engineer | MSc Candidate Cairo University

Highly skilled in applying AI solutions for real-world applications.

Experience in designing, developing, and deploying AI solutions for production environments.

WORK EXPERIENCE

• Senior AI Engineer

09/2023 – Present

Esbaar Company / Muscat, Oman

On-site

- Designed and deployed production-grade AI systems for safety and operational monitoring across industrial environments, focusing on real-time video analytics, reliability, and scalability.
- Architected and optimized high-performance inference pipelines using NVIDIA Triton Server and GStreamer, enabling concurrent processing of 50+ camera streams on commercial GPUs with dynamic batching and model parallelism.
- Accelerated inference efficiency by implementing Quantization-Aware Training (QAT) and INT8 post-training calibration with TensorRT, achieving significant latency and throughput improvements in production.
- Developed advanced computer vision models and algorithms for detecting, tracking, and associating personnel with their safety equipment, extending functionality to handle challenging visual conditions and complex environments.
- Built modular, containerized AI microservices with Docker and integrated distributed message streaming via Kafka, enabling fault-tolerant, event-driven communication and scalable orchestration exploration using Kubernetes.
- Implemented monitoring and reporting mechanisms within AI services to track data drift, pipeline performance, and per-stage latency, providing transparency and insights into real-time system behavior.
- Collaborated with frontend engineers to design and maintain OLTP and OLAP database architectures, ensuring optimized data flow between AI systems and analytical dashboards.
- Contributed to the development of an LLM-driven analytics assistant that converts natural language queries into SQL, powering dynamic dashboards and enabling intuitive interaction with structured data through Metabase.

• Reserved Temp. Officer

04/2021 - 10/2023

Mandatory Military Services in The Egyptian Armed Forces

On-site

• AI Research Assistant

02/2021 – 05/2021

OneLab Company / Dokki, Egypt

On-site

- Conducted applied research and model development for medical imaging, focusing on breast cancer detection, segmentation, and classification using deep learning and computer vision techniques.
- Designed and implemented a UNet-based segmentation model capable of identifying and localizing cancerous regions in X-ray images, achieving high precision and recall while differentiating between benign and malignant cases.
- Introduced additional morphological features, such as breast shape descriptors, to enhance model discriminative power and improve diagnostic accuracy beyond conventional architectures.
- Optimized and deployed deep learning models on FPGA hardware using the DPU framework and Xilinx Vitis Compiler; performed quantization (INT8) and calibration to maintain inference accuracy post-deployment.
- Generated custom FPGA-optimized model binaries (.xmodel) and integrated them into NI LabVIEW workflows, achieving efficient real-time execution on low-power edge devices.
- Contributed to the successful transition of the AI model from research to a market-ready medical solution, supporting deployment in clinical environments to assist radiologists in diagnosis and decision-making.

• Machine Learning Intern

01/2020 – 05/2020

Valeo Egypt / Smart Village

On-site

- Collaborated with Valeo's R&D team on developing deep learning models for autonomous driving perception, focusing on enhancing LiDAR-camera fusion for improved environmental understanding.
- Designed and implemented sensor fusion techniques to combine LiDAR point clouds with multi-view camera data, mitigating LiDAR sparsity and improving spatial consistency for downstream perception tasks.
- Applied advanced convolutional architectures and temporal modeling to increase the robustness of scene interpretation in varying road and weather conditions using the CARLA simulation environment.
- Contributed to the successful validation of fusion models that improved depth estimation and object detection accuracy in dynamic driving scenarios.
- Supported research collaboration between Cairo University and Valeo's R&D division; the work was later published in an academic conference (see Publications section).

• AI / Software Engineer Intern

07/2019 – 09/2019

SALEC Systems (Barq) Egypt / Maadi

On-site

- Developed an AI-driven employee attendance and movement analytics system using multi-camera computer vision pipelines to enhance workplace monitoring and performance insights.

- Designed and implemented a modular three-stage pipeline consisting of localization, detection, and cross-camera validation modules to ensure accurate and consistent identification.
- Built a face recognition framework using a Siamese neural network and few-shot learning for employee re-identification, enabling high-accuracy matching even with limited training samples.
- Engineered real-time detection and tracking algorithms optimized for multi-stream video input, achieving over 92% identification accuracy under varying environmental conditions.
- Collaborated with backend developers to integrate vision-based analytics with employee management databases, providing automated attendance tracking and behavioral statistics dashboards.

RESEARCH PUBLICATIONS

- **GraphDrive: Graph-Aided Decision-Making for Autonomous Vehicles** 2025
7th International Conference on Computer and Applications (IEEE-ICCA), Bahrain
 - Proposed **GraphDrive**, a GAT-based framework for interpretable decision-making in autonomous vehicles, modeling traffic entities as graph nodes with dynamic attention reasoning.
 - Achieved **14.6% higher decision accuracy** and **9.2% better path precision** on the DriveCoT benchmark, maintaining real-time inference (~ 15 FPS) with transparent attention-based reasoning.
 - Validated through extensive ablation studies showing optimal performance using a 12-layer GAT and 1024-dimensional embeddings; 87% of model errors were traceable via interpretable attention maps.
- **TimeFusionNet for End-to-End Self-Driving Cars** 2023
5th Novel Intelligent and Leading Emerging Sciences Conference (NILES), Egypt
 - Proposed **TimeFusionNet**, a temporal deep neural network with LiDAR-camera sensor fusion for end-to-end imitation learning in autonomous driving.
 - Captured temporal dependencies across frames using 3D-CNN and Voxel-based LiDAR modules, improving spatial awareness and decision stability.
 - Outperformed baseline models with **12–40% higher decision accuracy** on CoRL17 benchmarks, achieving real-time inference at ~ 10 FPS.

EDUCATION

- **MSc in Communications and Electronics Engineering** 2021 – 2025 (Expected)
Faculty of Engineering, Cairo University CGPA: 3.8/4.0
 - Master’s Thesis: *GraphDrive: Graph-Aided Decision-Making for Autonomous Vehicles*. Developed an interpretable decision-making framework using Graph Attention Networks (GATs) for autonomous driving. Accepted at **IEEE-ICCA 2025**.
- **BSc in Communications and Electronics Engineering** 2015 – 2020
Faculty of Engineering, Cairo University
 - Graduation Project: *High-Performance Deep Learning for Autonomous Vehicles*. Focused on developing advanced deep learning methods to enhance autonomous vehicle perception and control systems.
 - Designed a time-dependent deep neural network architecture for end-to-end driving, outperforming CoRL’17 benchmarks on Intel-based implementations.
 - Implemented LiDAR-camera fusion using VoxelNet to improve environmental perception and decision reliability.
 - Integrated reinforcement learning (Rainbow algorithm) with imitation learning in the CARLA simulator to refine driving behavior.
 - Recognized among the **Top 25 Graduate Projects in the Middle East** in Dell’s “Envision the Future” competition, supported by Valeo, ASRT, and ITAC.

TECHNICAL AND SOFT SKILLS

Programming Languages: Python, C/C++, SQL, Bash Scripting

AI and Deep Learning Frameworks: PyTorch, TensorFlow (Graph & Eager Execution), Keras, OpenCV, Hugging Face Transformers, LangChain, LangGraph

Model Architectures and Specializations: Convolutional Neural Networks (CNNs), 3D-CNNs, Graph Neural Networks (GNNs/GATs), Recurrent Neural Networks (RNNs, GRU, LSTM), UNet, Siamese Networks, Reinforcement Learning (Rainbow DQN), Generative AI, LLM-based Agents, RAG (Retrieval-Augmented Generation), and CAG (Chain-of-Attention Graphs)

Computer Vision and Perception: Object Detection (YOLO family), Image Segmentation, Activity Recognition, PPE Detection, Multi-Camera Tracking, Sensor Fusion (LiDAR-Camera), Edge AI Inference

Machine Learning & Data Science Tools: Scikit-Learn, Pandas, NumPy, Matplotlib, CVAT, Data Annotation & Preprocessing Pipelines, OLTP/OLAP Database Design

MLOps, Deployment & Optimization: NVIDIA Triton Inference Server, TensorRT, GStreamer, Docker, Microservices Architecture, Kubernetes, FastAPI, Django, MLFlow, Kafka, Model Quantization, Pruning, and Dynamic Batching

Cloud & DevOps Tools: AWS (EC2, S3), GCP, Git/GitHub, CI/CD Workflows, RESTful APIs, Edge and Cloud Deployment Pipelines

Research & Development Skills: Model Benchmarking, Explainable AI (XAI), Reproducible Experimentation, Academic Writing, and Paper Implementation (IEEE & NILES Published Work)

Soft Skills: Team Collaboration, Problem Solving, Rapid Learning, Research-oriented Thinking, Technical Communication, Project Ownership, Adaptability, and Mentorship

CERTIFICATIONS AND COURSES

• The Complete Agentic AI Engineering Course by Ed Donner	<i>Ongoing</i>
• AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents by Ed Donner	<u>Certified</u>
• Rapid Application Development with Large Language Models (LLMs) by Nvidia	<u>Certified</u>
• LangChain Chat with Your Data by deeplearning.ai	<u>Certified</u>
• Building Applications with Vector Databases by deeplearning.ai	<u>Certified</u>
• Building Real-Time Video AI Applications by Nvidia	<u>Certified</u>
• Getting Started with Google Kubernetes Engine by Coursera	<i>Ongoing</i>
• Fundamentals of AI Agents by HuggingFace	<u>Certified</u>
• Generative AI: Unleashing Next-Gen AI by Udemy	<u>Certified</u>
• Deep Learning Nanodegree by Udacity	<u>Certified</u>
• Retrieval augmented generation (RAG) Course by Langchain	<i>Online</i>
• Deep Learning Specialization Course by deeplearning.ai	<u>Certified</u>
• Machine Learning Course by Stanford University	<u>Certified</u>
• Convolutional Neural Networks by deeplearning.ai	<u>Certified</u>
• Custom Models, Layers, and Loss Functions with TensorFlow by deeplearning.ai	<u>Certified</u>
• Custom and Distributed Training with TensorFlow by deeplearning.ai	<u>Certified</u>
• Apply Generative Adversarial Networks (GANs) by deeplearning.ai	<u>Certified</u>
• Build Basic Generative Adversarial Networks (GANs) by deeplearning.ai	<u>Certified</u>
• Introduction to TensorFlow for AI, ML, and DL by deeplearning.ai	<u>Certified</u>
• Generative AI with Large Language Models by deeplearning.ai	<u>Certified</u>
• AI for everyone by deeplearning.ai	<u>Certified</u>
• Introduction to Git and GitHub by Google on Coursera	<i>Online</i>